ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PRACTICE EXAM 4 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics practice exam 4.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, sensible design choices and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 C ✓ 1.1.7 D ✓ 1.1.8 B ✓ 1.1.9 C ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 B ✓ 1.3.4 I ✓ 1.3.5 E ✓ 1.3.6 H ✓ 1.3.7 J ✓ 1.3.8 A ✓ 1.3.9 D ✓ 1.3.10 F ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 In order ✓✓✓✓✓: identify the problem (design brief); research / investigate; develop ideas and choose the best solution; plan and make a prototype; test and evaluate (and improve). (5)

2.2 Any FOUR ✓✓✓✓: moves a 300 kg gate 4,5 m; opens in less than about 25 s; works by remote control; stops at both ends; stops / reverses when the beam is broken; runs on battery during load-shedding; weatherproof; manual release. (4)

2.3 Inputs ✓✓: remote receiver, open and closed limit switches, safety beam (and push button). Controller ✓. Outputs ✓✓: relays / motor driver to the DC motor, warning light. Power supply / battery and correct arrows ✓ (6)

2.4 To warn people and drivers that the gate is moving so that nobody is hit or trapped ✓ (1)

2.5 Stop the motor immediately ✓ and then reverse / open the gate again (or wait until the beam is clear) ✓. (2)

[18]

QUESTION 3

3.1 $T = F \times r$ ✓ $= 400 \times 0{,}03$ ✓ $= 12\ N{\cdot}m$ ✓ (3)

3.2 $N = 1\,400 \div 20$ ✓ $= 70\ r/min$ ✓ (2)

3.3 $v = \pi dN \div 60$ ✓ $= \pi \times 0{,}06 \times 70 \div 60$ ✓ $= 0{,}22\ m/s$ ✓ (3)

3.4 $t = s \div v$ ✓ $= 4{,}5 \div 0{,}22 = 20{,}47\ s$ ✓ (accept 20,45 s with 0,22) (2)

3.5 $P = F \times v \checkmark = 400 \times 0{,}22 \checkmark = 87{,}96\ W\ (\approx 88\ W) \checkmark$ (3)

3.6 $T_{motor} = 12 \div 20 \checkmark = 0{,}6\ N{\cdot}m \checkmark$ (2)

3.7 The motor turns too fast (1 400 r/min) for the gate ✓; the gearbox lowers the speed to a safe gate speed ✓ and increases the torque so that a small motor can move the heavy gate ✓. (3)

[18]

QUESTION 4

4.1 Controller output (through a transistor) to the relay coil ✓; flywheel diode across the coil ✓; 12 V supply ✓; fuse in series ✓; relay contacts in series with the motor ✓; correct symbols, neat, labelled ✓ (6)

4.2 Reverse the polarity of the supply to the motor ✓ using two relays (one for each direction) or a DPDT relay / H-bridge ✓, with an interlock so that both directions can never be on together ✓. (3)

4.3 A limit switch is fitted at each end of the track ✓; when the gate reaches it, the switch changes state ✓ and the controller (or the switch in the circuit) switches off the motor for that direction ✓. (3)

4.4 $\text{Charge per cycle} = 8 \times (20 + 20) = 320\ As \checkmark = 320 \div 3\,600 = 0{,}089\ Ah \checkmark;$ $\text{cycles} = 7 \div 0{,}089 \checkmark = 78\ \text{complete cycles} \checkmark$ (4)

4.5 10 A ✓ – it carries the normal 8 A without blowing but blows quickly on a fault (5 A would blow in normal use; 30 A would not protect the motor) ✓ (2)

[18]

QUESTION 5

5.1 Any FOUR ✓✓✓✓: infrared safety beam; obstacle / current sensing that stops or reverses the gate; flashing warning light; manual release; limit switches; anti-crush rubber edge; overload protection; secure remote codes. (4)

5.2 (a) galvanised steel (or aluminium) ✓ – strong and resists rust outdoors ✓ (b) nylon or steel rack ✓ – nylon is quiet and does not rust; steel is very strong for heavy gates ✓ (4)

5.3 Any FOUR ✓✓✓✓: clean the track and remove stones; check and clean the safety-beam sensors; check rack and pinion for wear and alignment; test the back-up battery; test the limit switches and manual release; tighten bolts; lubricate moving parts as recommended. (4)

5.4 Any FOUR ✓✓✓✓: obstruction on the track; safety beam blocked or dirty; flat battery / power failure; overload tripped because the gate is stiff; faulty or loose wiring; pinion slipping off the rack; a limit switch faulty or out of position. (4)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE